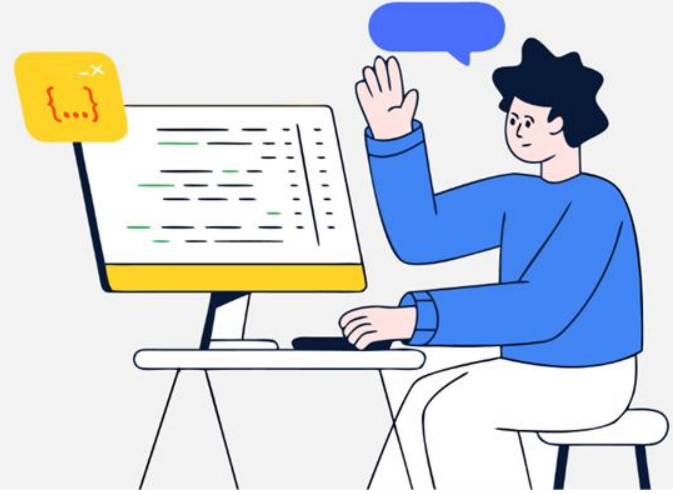




Solution Challenge



Team Details

- Team name:** The Pythonic Four
- Team leader name:** Yashraj Mishra
- Problem Statement:** *A full-stack, QR code and geolocation-based attendance system that prevents proxy attendance by ensuring students are physically present within a faculty-defined classroom radius before marking attendance*

Brief about your solution:

Our solution, **Attendify**, is a full-stack, geolocation-based attendance management system designed to completely eliminate proxy attendance in educational institutions. The core idea is simple yet powerful — a student can only mark their attendance if they are **physically present** in the classroom, and we enforce this through a **dual-verification mechanism** combining dynamic QR codes and real-time GPS validation.

Here's how it works: the faculty logs into their dashboard and creates an attendance session by capturing their **live GPS coordinates** and setting a **proximity radius** — say 50 meters. The system then generates a **unique, session-specific QR code** which the faculty projects on the classroom screen. Students open the app, scan the QR code, and the system simultaneously validates two things — first, that the QR token is **valid and active**, and second, that the student's **live location falls within the defined radius**. If either check fails, attendance is not marked. There is no way to mark attendance remotely.

On the management side, faculty can **bulk upload student data via Excel**, view live active sessions, and **export attendance reports as CSV** for records and analysis. Students get a clean dashboard to view their full attendance history. The entire system is secured with **JWT-based authentication and BCrypt password hashing**, ensuring data integrity and role-based access control throughout.

Built on a **React + Node.js + SQLite** stack, Attendify is lightweight, production-ready, and requires zero database configuration — making it easily deployable for any institution looking to modernize their attendance infrastructure.

Opportunities

- a. How different is it from any of the other existing ideas?
- b. How will it be able to solve the problem?
- c. USP of the proposed solution

Most existing attendance systems — like biometric devices, RFID cards, or simple mobile check-ins — can be **spoofed, shared, or operated remotely**. Apps like Google Forms or basic QR attendance tools verify the QR but have **no location enforcement**, meaning a student can share a screenshot of the QR and a friend marks attendance from home. Biometric systems are **hardware-dependent and expensive** to install in every classroom. Attendify is different because it enforces **simultaneous dual verification** — a valid session token AND physical presence — without requiring any additional hardware. It runs entirely on a smartphone and a browser, making it **zero-infrastructure and scalable**.

Attendify tackles proxy attendance at its root by making it **technically impossible** to mark attendance without being present. The dynamic QR code changes with every session so old codes are useless. The geolocation check runs in real-time on the student's device and cross-validates with the faculty's set coordinates on the server. Even if a student shares the QR code with someone outside, that person **cannot pass the location check** from a different place. The faculty has complete visibility through live session tracking and downloadable reports, closing the loop on attendance integrity.

Attendify's biggest differentiator is its dual verification approach — a student must simultaneously hold a valid QR session token AND be physically within the classroom's GPS radius to mark attendance. This makes proxy attendance ****technically impossible****, not just discouraged. On top of that, it requires ****zero additional hardware****, runs entirely on a smartphone, and is deployable in minutes — giving institutions enterprise-grade attendance security at virtually no infrastructure cost.

List of features offered by the solution

Geolocation-based attendance with custom classroom radius

Dynamic QR code generation per session

Dual verification — QR token + live GPS simultaneously

Faculty dashboard for session creation and management

One-click CSV export for attendance reports

Bulk student upload via Excel

Student portal with QR scanner and attendance history

JWT authentication with role-based access control

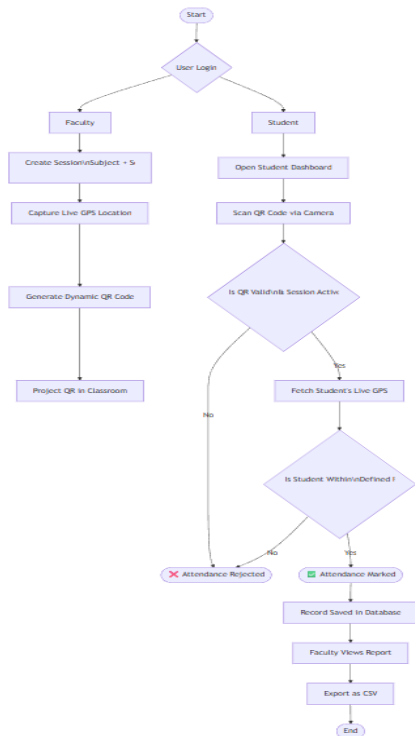
BCrypt password hashing for secure data storage

Auto-expiring session tokens to prevent QR reuse

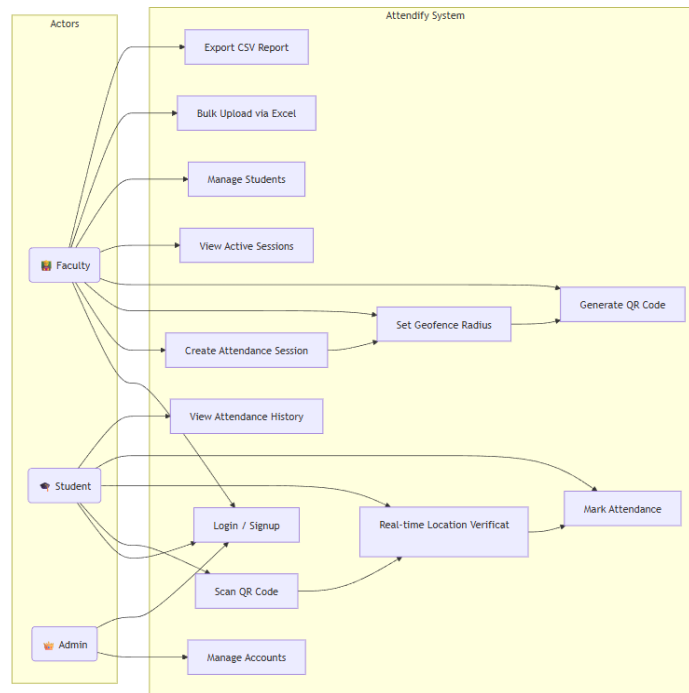
Admin panel for platform-wide account management

Process flow diagram or Use-case diagram

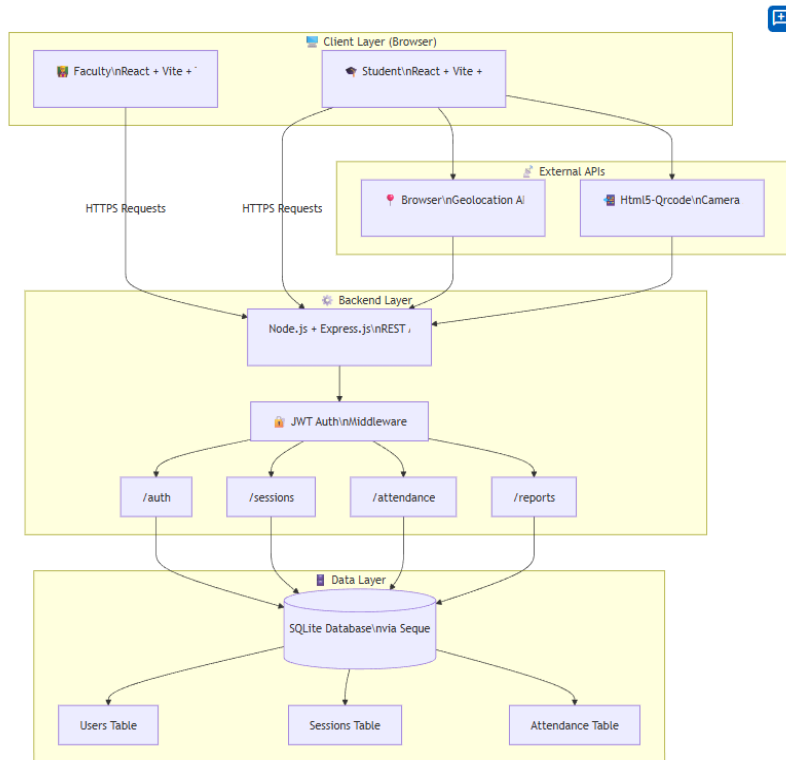
Process Flow Diagram



Use Case Diagram



Architecture diagram of the proposed solution



Technologies to be used in the solution

React.js (Vite) — Frontend UI framework

Tailwind CSS — Responsive styling

Html5-Qrcode — In-browser QR code scanning

Node.js — Backend runtime environment

Express.js — REST API framework

SQLite + Sequelize ORM — Lightweight database

JWT — Secure user authentication

BCrypt — Password hashing & encryption

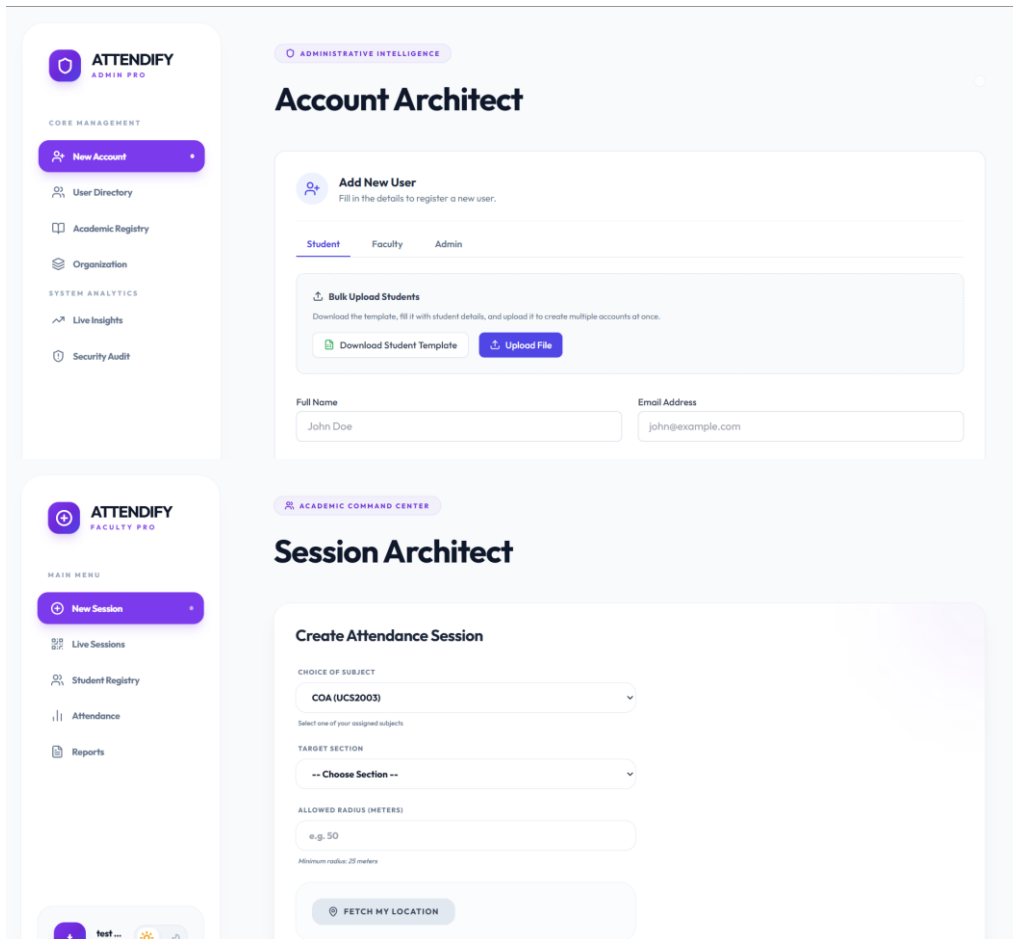
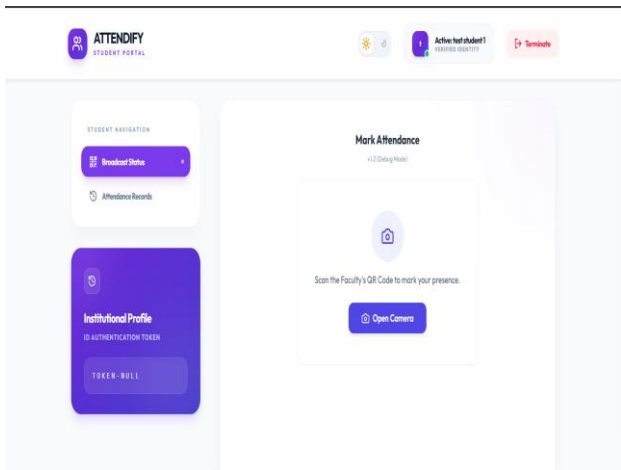
Browser Geolocation API — Real-time GPS tracking

Multer + XLSX — Excel file upload & parsing

Axios — Frontend-backend API communication

Render + Firebase — Backend & frontend deployment

Snapshots of the MVP



Additional Details/Future Development (if any)

- Dedicated Android/iOS mobile app using React Native
- AI-based anomaly detection to flag suspicious attendance patterns
- Advanced analytics with trend graphs and performance heatmaps
- Push notifications for session alerts and low attendance warnings
- Multi-institution / multi-tenant support for multiple colleges
- Timetable integration for auto-session creation
- HOD-level dashboard for department-wide attendance overview
- Offline mode with sync when internet is restored
- ERP integration with university systems like Moodle/Fedena
- Automated weekly/monthly attendance email reports

Provide links to your:

1. GitHub Public Repository:

<https://github.com/Yashrajmshr/Attendify/>

2. Demo Video Link (3 Minutes)

<https://drive.google.com/drive/folders/120uJqtLd2QWI41kzIT1hZPLyHIEO8w7u?dmr=1&ec=wgcdrive-%5Bmodule%5D-goto>

3. MVP Link

<https://drive.google.com/drive/folders/120uJqtLd2QWI41kzIT1hZPLyHIEO8w7u?dmr=1&ec=wgcdrive-%5Bmodule%5D-goto>

4. Working Prototype Link <https://attendify-frontend-lh87.onrender.com/login>

For Admin Login Use: test_admin@gmail.com

password: 1234

For Faculty Login Use: test_facult1@gmail.com

password: 1234

For Student Login use: test_student1@gmail.com

password: 1234



Solution Challenge



Thank You